

Apple

As a Data Analyst on the Supply Chain Procurement Team, you are tasked with assessing supplier performance to ensure reliable delivery of critical components....

Question 1: We need to know who our most active suppliers are. Identify the top 5 suppliers based on the total volume of components delivered in October 2024.

```
SELECT
    s.supplier_name AS supplier_name,
    SUM(sd.component_count) AS total_volumn
FROM supplier_deliveries AS sd
LEFT JOIN suppliers AS s
    ON sd.supplier_id = s.supplier_id
WHERE delivery_date BETWEEN '2024-10-01' AND '2024-10-31'
GROUP BY s.supplier_name
ORDER BY total_volumn DESC
LIMIT 5;
```



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Question 2: For each region, find the supplier ID that delivered the highest number of components in November 2024. This will help us understand which supplier is handling the most volume per market.

```
WITH RankedSuppliers AS (  
    SELECT  
        sd.manufacturing_region,  
        sd.supplier_id,  
        SUM(sd.component_count) AS total_volume,  
        ROW_NUMBER() OVER (  
            PARTITION BY sd.manufacturing_region  
            ORDER BY SUM(sd.component_count) DESC  
        ) AS region_rank  
    FROM supplier_deliveries AS sd  
    WHERE sd.delivery_date BETWEEN '2024-11-01' AND '2024-11-30'  
    GROUP BY sd.manufacturing_region, sd.supplier_id  
)  
  
SELECT  
    manufacturing_region,  
    supplier_id,  
    total_volume  
FROM RankedSuppliers  
WHERE region_rank = 1  
ORDER BY manufacturing_region;
```



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Question 3: We need to identify potential gaps in our supply chain for Asia. List all suppliers by name who have not delivered any components to the 'Asia' manufacturing region in December 2024.

```
SELECT
    s.supplier_name
FROM suppliers AS s
WHERE NOT EXISTS (
    SELECT 1
    FROM supplier_deliveries AS sd
    WHERE sd.supplier_id = s.supplier_id
        AND sd.manufacturing_region = 'Asia'
        AND sd.component_count > 0
        AND sd.delivery_date BETWEEN '2024-12-01' AND '2024-12-31'
)
ORDER BY s.supplier_name;
```

